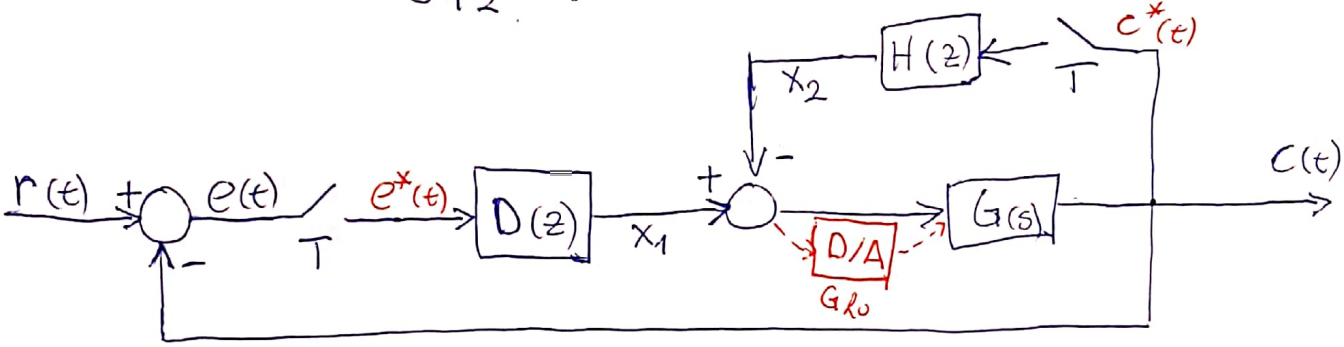
15.04.2019. K1

1. Импулсни систем управљања дат је на слици. $D(z)$ ради по алгоритму $x_1(kT) = 2e(kT)$, док $H(z)$ ради по алгоритму $x_2(kT) = 3c(kT)$. $G(s)$ је представљен са $G(s) = \frac{1}{s+2}$. Усвојена је периода одмјеравања $T = 1[s]$.



a) Одредити ф-ју дискретног преноса $\frac{C(z)}{R(z)}$

b) Одредити импулсни одзив система.

RJEŠENJE:

- Испред блока $G(s)$ потребно је поставити D/A конвертор (кола задршке нултог реда)

$$G_{h0}(s) = \frac{1-e^{-sT}}{s}$$

a) $C(s) = G_{h0}(s) \cdot G(s) \left[X_1(s) - X_2(s) \right]$

$$X_1(s) = D(z) \cdot E^*(s) \xrightarrow[\text{ТАФДП}]{} X_1^*(s) = D(z) \cdot E^*(s) \,/\, Z \Rightarrow X_1(z) = D(z) \cdot E(z)$$

$$X_2(s) = H(z) \cdot C^*(s) \xrightarrow[\text{ТАФДП}]{} X_2^*(s) = H(z) \cdot C^*(s) \,/\, Z \Rightarrow X_2(z) = H(z) \cdot C(z)$$

$$D(z): \; x_1(kT) = 2e(kT) \,/\, Z \Rightarrow X_1(z) = 2E(z) \Rightarrow D(z) = 2$$

$$H(z): \; x_2(kT) = 3c(kT) \,/\, Z \Rightarrow X_2(z) = 3C(z) \Rightarrow H(z) = 3$$

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$$X_1(s) = D(z)\cdot E^*(s) = 2E^*(s)$$

$$X_2(s) = H(z)\cdot C^*(s) = 3\,C^*(s)$$

$$C(s) = G_{\ell_0}(s)\cdot G(s)\cdot\left[X_1(s) - X_2(s)\right]$$

$$C(s) = 2\,G_{\ell_0}(s)\cdot G(s)\cdot E^*(s) - 3\,G_{\ell_0}(s)\cdot G(s)\cdot C^*(s)$$

$$E(s) = R(s) - C(s) \underset{\text{ТАФДП}}{\Longrightarrow} E^*(s) = R^*(s) - C^*(s)$$

$$C(s) = 2\,G_{\ell_0}(s)\cdot G(s)\cdot R^*(s) - 2\,G_{\ell_0}(s)\cdot G(s)\cdot C^*(s) - 3\,G_{\ell_0}(s)\cdot G(s)\cdot C^*(s)$$

$$C(s) = 2\,G_{\ell_0}(s)\cdot G(s)\cdot R^*(s) - 5\,G_{\ell_0}(s)\cdot G(s)\cdot C^*(s) \underset{\text{ТАФДП}}{\Longrightarrow}$$

$$C^*(s) = 2\,G_{\ell_0}G^*(s)\cdot R^*(s) - 5\,G_{\ell_0}G^*(s)\cdot C^*(s) \quad /z$$

$$C(z)\left[1 + 5\,G_{\ell_0}G(z)\right] = 2\,G_{\ell_0}G(z)\cdot R(z)$$

$$\Rightarrow \boxed{\frac{C(z)}{R(z)} = \frac{2\,G_{\ell_0}G(z)}{1 + 5\,G_{\ell_0}G(z)}}$$

$$G_{\ell_0}G(z) = Z\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s+2}\right\} = (1-z^{-1})\,Z\left\{\frac{1}{s(s+2)}\right\}$$

$$\frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} = \frac{As + 2A + Bs}{s(s+2)} \Rightarrow \begin{array}{l} A + B = 0 \\ 2A = 1 \end{array} \Rightarrow A = \frac{1}{2},\ B = -\frac{1}{2}$$

$$G_{\ell_0}G(z) = (1-z^{-1})\,Z\left\{\frac{1/2}{s} - \frac{1/2}{s+2}\right\} = \frac{1}{2}\cdot(1-z^{-1})\,Z\left\{\frac{1}{s} - \frac{1}{s+2}\right\}$$

$$G_{\ell_0}G(z) = \frac{1}{2}\cdot\frac{z-1}{z}\cdot\left[\frac{z}{z-1} - \frac{z}{z-e^{-2T}}\right], \quad T = 1\,[s]$$

$$G_{\ell_0}G(z) = \frac{1}{2}\cdot\left[1 - \frac{(z-1)}{(z-0.1353)}\right] = \frac{1}{2}\cdot\frac{\cancel{z} - 0.1353 - \cancel{z} + 1}{(z-0.1353)}$$

$$G_{\ell_0}G(z) = \frac{0.43235}{(z-0.1353)}$$

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$$\frac{C(z)}{R(z)} = \frac{2 \cdot \frac{0.43236}{(z-0.1353)}}{1 + 5 \cdot \frac{0.43235}{(z-0.1353)}} = \frac{\frac{0.8647}{\cancel{(z-0.1353)}}}{\frac{z-0.1353+2.16175}{\cancel{(z-0.1353)}}}$$

$$\boxed{\frac{C(z)}{R(z)} = \frac{0.8647}{(z+2.02645)}}$$

b) Импулсни одзив $r(kT) = \delta(kT)/z \Rightarrow R(z) = 1$

$$C(z) = \frac{0.8647}{(z+2.02645)} \cdot R(z)$$

$$\Rightarrow C(z) = \frac{0.8647}{(z+2.02645)}$$

$$c(kT) = Z^{-1}\{C(z)\}$$

$$c(kT) = 0.8647\, Z^{-1}\left\{\frac{1}{z+2.02645}\right\} = 0.8647\, Z^{-1}\left\{\frac{z^{-1} \cdot z}{z+2.02645}\right\}$$

$$\boxed{c(kT) = 0.8647 \cdot 2.02645^{(k-1)T}}$$

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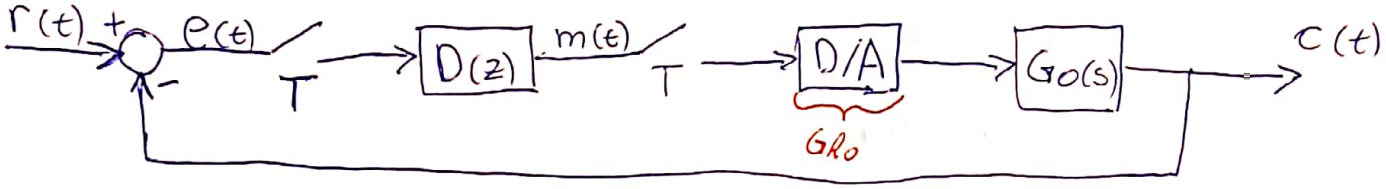
2. Систем аутоматског управљања дат је блок дијаграмом на слици. $D(z)$ ради по алгоритму $m(k) = 2e(k)$. Објекат управљања описан је са $G_o(s) = \frac{1}{s(s+2)}$. Усвојена је периода одмјеравања $T = 1[s]$.



a) Одредити вриједности сигнала $c(t)$ у тренуцима $t = 1.1T$ и $t = 2.1T$ ако се на улаз доведе јединични одскочни сигнал.

b) Одредити тачност рада система у стационарном стању ако се на улаз доведе сигнал $r(t) = 5 + 2t$.

РЈЕШЕЊЕ:

a) Када је потребно одредити вриједност сигнала између тренутака одабирања користи се модификована Z трансформација (z_m)

$$z_m^{-1}\{F(z,m)\} = f[kT - (1-m)T]$$

~~Нама~~ Нама треба $t = 1.1T$ и $t = 2.1T \Rightarrow m = 0.1$

$$C(s) = G_{Ro}(s) \cdot G_o(s) \cdot M^*(s) \xrightarrow[\text{ТАФДП}]{} C^*(s) = G_{Ro}G_o^*(s) \cdot M^*(s)$$

$$M(s) = D(z) \cdot E^*(s) \xrightarrow[\text{ТАФДП}]{} M^*(s) = D(z) \cdot E^*(s)$$

$$E(s) = R(s) - C(s) \xrightarrow[\text{ТАФДП}]{} E^*(s) = R^*(s) - C^*(s)$$

$$M^*(s) = D(z) \cdot E^*(s) = D(z) \cdot R^*(s) - D(z) \cdot C^*(s)$$

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$$C^*(s) = G_{h_0}G_{0(s)}^* \cdot M^*(s) = G_{h_0}G_{0(s)}^* \cdot D(z) \cdot R^*(s) - G_{h_0}G_{0(s)}^* \cdot D(z) \cdot C^*(s)$$

$$C^*(s)\left[1 + G_{h_0}G_{0(s)}^* \cdot D(z)\right] = G_{h_0}G_{0(s)}^* \cdot D(z) \cdot R^*(s)$$

$$C^*(s) = \frac{G_{h_0}G_{0(s)}^* \cdot D(z)}{1 + G_{h_0}G_{0(s)}^* \cdot D(z)} \cdot R^*(s) \Big/ \cdot e^{-(1-m)sT}$$

→ јер нам треба Z_m

$$\left[C(s) \cdot e^{-(1-m)sT}\right]^* = \frac{\left[G_{h_0}G_{0(s)} \cdot e^{-(1-m)sT}\right]^* \cdot D(z)}{1 + G_{h_0}G_{0(s)}^* \cdot D(z)} \cdot R^*(s) \Big/ Z$$

$$C(z,m) = \frac{G_{h_0}G_{0(z,m)} \cdot D(z)}{1 + G_{h_0}G_{0(z)} \cdot D(z)} \cdot R(z)$$

$$D(z): \quad m(k) = 2e(k) \,/ Z \Rightarrow \left.\begin{array}{l} M(z) = 2E(z) \\ M(z) = D(z) \cdot E(z) \end{array}\right\} \Rightarrow D(z) = 2$$

$$r(t) = h(t) \,/ Z \Rightarrow R(z) = \frac{z}{z-1}$$

→ улаз је јединични одскочни сигнал

$$G_{h_0}G_0(z) = Z\left\{\frac{1-e^{-sT}}{s} \cdot \frac{1}{s(s+2)}\right\} = (1-z^{-1})\, Z\left\{\frac{1}{s^2(s+2)}\right\}$$

$$\frac{1}{s^2(s+2)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+2} = \frac{As + 2A + Bs^2 + 2Bs + Cs^2}{s^2(s+2)}$$

$$\left.\begin{array}{r} B + C = 0 \\ A + 2B = 0 \\ 2A = 1 \end{array}\right\} \Rightarrow A = \frac{1}{2}, \; B = -\frac{1}{4}, \; C = \frac{1}{4}$$

$$G_{h_0}G_0(z) = (1-z^{-1})\, Z\left\{\frac{1/2}{s^2} - \frac{1/4}{s} + \frac{1/4}{s+2}\right\}$$

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$$G_{h_0}G_0(z) = \frac{1}{4}(1-z^{-1})\,\mathcal{Z}\left\{\frac{2}{s^2} - \frac{1}{s} + \frac{1}{s+2}\right\}$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{2Tz}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-2T}}\right], \quad T=1[s]$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{2z}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-0.1353}\right]$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\left[\frac{2}{(z-1)} - 1 + \frac{z-1}{(z-0.1353)}\right]$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\frac{2(z-0.1353)-(z-1)(z-0.1353)+(z-1)^2}{(z-1)\,(z-0.1353)}$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\frac{\cancel{2z}-0.2706\cancel{-z^2}+1.1353z-0.1353\cancel{+z^2}\cancel{-2z}+1}{(z-1)(z-0.1353)}$$

$$G_{h_0}G_0(z) = \frac{1}{4}\cdot\frac{1.1353z+0.5941}{(z-1)(z-0.1353)} \Rightarrow \underline{\underline{G_{h_0}G_0(z) = \frac{0.2838z+0.1485}{(z-1)(z-0.1353)}}}$$

$$G_{h_0}G_0(z,m) = \mathcal{Z}_m\left\{\frac{1-e^{-sT}}{s}\cdot\frac{1}{s(s+2)}\right\} = (1-z^{-1})\,\mathcal{Z}_m\left\{\frac{1}{s^2(s+2)}\right\}$$

$$G_{h_0}G_0(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\,\mathcal{Z}_m\left\{\frac{2}{s^2} - \frac{1}{s} + \frac{1}{s+2}\right\}$$

$$G_{h_0}G_0(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{2mT}{z-1} + \frac{2T}{(z-1)^2} - \frac{1}{z-1} + \frac{e^{-2mT}}{z-e^{-2T}}\right]$$

$T=1[s]$, $m=0.1$

$$G_{h_0}G_0(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{0.2}{z-1} + \frac{2}{(z-1)^2} - \frac{1}{z-1} + \frac{e^{-0.2}}{z-e^{-2}}\right]$$

$$G_{h_0}G_0(z,m) = \frac{1}{4}\cdot\frac{z-1}{z}\cdot\left[\frac{2}{(z-1)^2} - \frac{0.8}{(z-1)} + \frac{0.8187}{(z-0.1353)}\right]$$

$$G_{h_0}G_0(z,m) = \frac{z-1}{z}\cdot\left[\frac{0.5}{(z-1)^2} - \frac{0.2}{(z-1)} + \frac{0.2047}{(z-0.1353)}\right]$$

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$$G_{h_0}G_0(z,m) = \frac{\cancel{z-1}}{z} \cdot \frac{0.5(z-0.1353) - 0.2\overbrace{(z-1)(z-0.1353)}^{z^2-1.1353z+0.1353} + 0.2047(z-1)^2}{(z-1)^{\cancel{2}}(z-0.1353)}$$

$$G_{h_0}G_0(z,m) = \frac{0.5z - 0.06765 - 0.2z^2 + 0.22706z - 0.02706 + 0.2047z^2 - 0.4094z + 0.2047}{z(z-1)(z-0.1353)}$$

$$G_{h_0}G_0(z,m) = \frac{0.0047z^2 + 0.31766z + 0.10999}{z(z-1)(z-0.1353)}$$

$$C(z,m) = \frac{G_{h_0}G_0(z,m)\cdot D(z)}{1 + G_{h_0}G_0(z)\cdot D(z)} \cdot R(z)$$

$$C(z,m) = \frac{\dfrac{0.0047z^2 + 0.31766z + 0.10999}{z(z-1)(z-0.1353)} \cdot 2}{1 + \dfrac{0.2838z + 0.1485}{(z-1)(z-0.1353)} \cdot 2} \cdot \frac{z}{z-1}$$

$$C(z,m) = \frac{\dfrac{0.0094z^2 + 0.63532z + 0.21998}{\cancel{z(z-1)(z-0.1353)}}}{\dfrac{z^2 - 1.1353z + 0.1353 + 0.5676z + 0.2970}{\cancel{(z-1)(z-0.1353)}}} \cdot \frac{\cancel{z}}{z-1}$$

$$C(z,m) = \frac{0.0094z^2 + 0.63532z + 0.21998}{z^2 - 0.5677z + 0.4323} \cdot \frac{1}{z-1}$$

$$C(z,m) = \frac{0.0094z^2 + 0.63532z + 0.21998}{z^3 - z^2 - 0.5677z^2 + 0.5677z + 0.4323z - 0.4323}$$

$$C(z,m) = \frac{0.0094z^2 + 0.63532z + 0.21998}{z^3 - 1.5677z^2 + z - 0.4323}$$

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$$c[kT-(1-m)T] = Z_m^{-1}\{C(z,m)\}$$

$$(0.0094z^2+0.63532z+0.21998):(z^3-1.5677z^2+z-0.4323) = 0.0094z^{-1} + 0.6501z^{-2} + 1.2298 + \dots$$

$$-(0.0094z^2-0.0147z+0.0094-0.0041z^{-1})$$

$$0.6501z + 0.2106 + 0.0041z^{-1}$$

$$-(0.6501z-1.0192+0.6501z^{-1}-0.2810z^{-2})$$

$$1.2298-0.6460z^{-1}+0.2810z^{-2}$$

$$\vdots$$

$$k=2 \Rightarrow c[2T-(1-0.1)T] = c[1.1T] = 0.6501$$

$$k=3 \Rightarrow c[3T-(1-0.1)T] = c[2.1T] = 1.2298$$

b)

- За анализу тачности рада система у стационарном стању користи се ф-ја повратног дискретног преноса (ф-ја преноса директне гране)

$$W(z) = G_{h_0}G_0(z)\cdot D(z) = \frac{2(0.2838z+0.1485)}{(z-1)(z-0.1353)}$$

- Улазни сигнал је задан као збир одскочне и линеарне ф-је ($r(t)=5+2t$), па се грешка у стационарном стању добија као збир грешака када на систем дејствује само одскочни референтни сигнал и када дејствује само линеарна ф-ја као референтни улаз

$$e(\infty) = \frac{r_0}{1+K_p} + \frac{vT}{K_v}$$

, K_p - константа положаја, K_v - брзинска константа

$$r(t) = 5+2t \Rightarrow r_0 = 5,\ v = 2$$

$$K_p = \lim_{z\to 1} W(z) = \lim_{z\to 1} \frac{2(0.2838z+0.1485)}{(z-1)(z-0.1353)} = \infty$$

$$K_v = \lim_{z\to 1}(z-1)W(z) = \lim_{z\to 1}\cancel{(z-1)}\cdot\frac{2\cdot(0.2838z+0.1485)}{\cancel{(z-1)}(z-0.1353)} = 1$$

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$$e(\infty) = \frac{r_0}{1+k_p} + \frac{vT}{k_v}, \quad T = 1[s]$$

$$e(\infty) = \frac{5}{1+\infty} + \frac{2\cdot 1}{1} \Rightarrow \boxed{e(\infty) = 2}$$

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3. Систем аутоматског управљања је описан моделом у простору стања. Усвојена је период одмјеравања $T = 1[s]$.

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -14 & -9 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 7 & 0 \end{bmatrix} x(t)$$

a) Одредити модел система у дискретном простору стања.

b) Одредити ф-ју дискретног преноса система.

РЈЕШЕЊЕ:

a) $x(k+1) = E\,x(k) + F\,u(k)$

$y(k) = C\,x(k)$

$$E(T) = e^{AT} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -14 & -9 \end{bmatrix}\right)^{-1}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\begin{bmatrix} s & -1 \\ 14 & s+9 \end{bmatrix}^{-1}\right\}\Big|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s^2+9s+14}\begin{bmatrix} s+9 & 1 \\ -14 & s \end{bmatrix}\right\}\Big|_{t=T} = \quad (s^2+9s+14) = (s+2)(s+7)$$

$$= \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{s+9}{(s+2)(s+7)} & \frac{1}{(s+2)(s+7)} \\ \frac{-14}{(s+2)(s+7)} & \frac{s}{(s+2)(s+7)} \end{bmatrix}\right\}\Big|_{t=T}$$

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$$\frac{s+9}{(s+2)(s+7)} = \frac{A}{s+2} + \frac{B}{s+7} = \frac{As+7A+Bs+2B}{(s+2)(s+7)}$$

$$\left.\begin{array}{l} A+B=1 \;/\cdot(-7) \\ 7A+2B=9 \end{array}\right\} \Rightarrow \left.\begin{array}{l} -7A-7B=-7 \\ 7A+2B=9 \end{array}\right\} \Rightarrow \begin{array}{l} -5B=2 \\ B=-\frac{2}{5}, \; A=\frac{7}{5} \end{array}$$

$$\frac{1}{(s+2)(s+7)} = \frac{A}{s+2} + \frac{B}{s+7} = \frac{As+7A+Bs+2B}{(s+2)(s+7)}$$

$$\left.\begin{array}{l} A+B=0 \Rightarrow A=-B \\ 7A+2B=1 \end{array}\right\} \Rightarrow 5A=1 \Rightarrow A=\frac{1}{5}, \; B=-\frac{1}{5}$$

$$\frac{-14}{(s+2)(s+7)} = \frac{A}{s+2} + \frac{B}{s+7} = \frac{As+7A+Bs+2B}{(s+2)(s+7)}$$

$$\left.\begin{array}{l} A+B=0 \Rightarrow A=-B \\ 7A+2B=-14 \end{array}\right\} \Rightarrow 5A=-14 \Rightarrow A=-\frac{14}{5}, \; B=\frac{14}{5}$$

$$\frac{s}{(s+2)(s+7)} = \frac{A}{s+2} + \frac{B}{s+7} = \frac{As+7A+Bs+2B}{(s+2)(s+7)}$$

$$\left.\begin{array}{l} A+B=1 \;/\cdot(-2) \\ 7A+2B=0 \end{array}\right\} \Rightarrow \left.\begin{array}{l} -2A-2B=-2 \\ 7A+2B=0 \end{array}\right\} \Rightarrow 5A=-2 \Rightarrow \begin{array}{l} A=-\frac{2}{5} \\ B=\frac{7}{5} \end{array}$$

$$E(T)=\mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{7/5}{s+2}-\frac{2/5}{s+7} & \frac{1/5}{s+2}-\frac{1/5}{s+7} \\ \frac{-14/5}{s+2}+\frac{14/5}{s+7} & \frac{-2/5}{s+2}+\frac{7/5}{s+7} \end{bmatrix}\right\}\Bigg|_{t=T} =$$

$$=\frac{1}{5}\mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{7}{s+2}-\frac{2}{s+7} & \frac{1}{s+2}-\frac{1}{s+7} \\ \frac{-14}{s+2}+\frac{14}{s+7} & \frac{-2}{s+2}+\frac{7}{s+7} \end{bmatrix}\right\}\Bigg|_{t=T} =$$

$$=\frac{1}{5}\begin{bmatrix} 7e^{-2t}-2e^{-7t} & e^{-2t}-e^{-7t} \\ 14(-e^{-2t}+e^{-7t}) & -2e^{-2t}+7e^{-7t} \end{bmatrix}\Bigg|_{t=T}$$

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$$E(T) = \frac{1}{5}\begin{bmatrix} 7e^{-2T} - 2e^{-7T} & e^{-2T} - e^{-7T} \\ 14(-e^{-2T} + e^{-7T}) & -2e^{-2T} + 7e^{-7T} \end{bmatrix}, \quad T = 1[s]$$

$$E(T) = \frac{1}{5}\begin{bmatrix} 0.9455 & 0.1344 \\ -1.8819 & -0.2643 \end{bmatrix} \Rightarrow \underline{\underline{E(T) = \begin{bmatrix} 0.1891 & 0.0269 \\ -0.3764 & -0.0529 \end{bmatrix}}}$$

$$F(T) = \int_0^T e^{At} B\,dt \cancel{= \frac{1}{5}\int_0^T \begin{bmatrix} 0.18 \end{bmatrix}}$$

$$F(T) = \frac{1}{5}\int_0^T \begin{bmatrix} 7e^{-2t} - 2e^{-7t} & e^{-2t} - e^{-7t} \\ 14(-e^{-2t} + e^{-7t}) & -2e^{-2t} + 7e^{-7t} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} dt$$

$$F(T) = \frac{1}{5}\int_0^T \begin{bmatrix} e^{-2t} - e^{-7t} \\ -2e^{-2t} + 7e^{-7t} \end{bmatrix} dt = \frac{1}{5}\begin{bmatrix} -\frac{1}{2}e^{-2t} + \frac{1}{7}e^{-7t} \\ e^{-2t} - e^{-7t} \end{bmatrix}\Bigg|_0^T$$

$$F(T) = \frac{1}{5}\begin{bmatrix} -\frac{1}{2}e^{-2T} + \frac{1}{7}e^{-7T} + \frac{1}{2}e^{-2\cdot 0} - \frac{1}{7}e^{-7\cdot 0} \\ e^{-2T} - e^{-7T} - e^{-2\cdot 0} + e^{-7\cdot 0} \end{bmatrix}, \quad T = 1[s]$$

$$F(T) = \frac{1}{5}\begin{bmatrix} 0.2896 \\ 0.1344 \end{bmatrix} \Rightarrow \underline{\underline{F(T) = \begin{bmatrix} 0.05792 \\ 0.02688 \end{bmatrix}}}$$

$$\boxed{\begin{aligned} X(k+1) &= EX(k) + Fu(k) = \begin{bmatrix} 0.1891 & 0.0269 \\ -0.3764 & -0.0529 \end{bmatrix} X(k) + \begin{bmatrix} 0.05792 \\ 0.02688 \end{bmatrix} u(k) \\ Y(k) &= CX(k) = \begin{bmatrix} 7 & 0 \end{bmatrix} X(k) \end{aligned}}$$

дискретан модел система у простору стања

b)

$$X(k+1) = EX(k) + Fu(k) \,/\, Z$$

$$zX(z) = EX(z) + FU(z)$$

$$(zI-E)^{-1} \cdot / (zI - E)X(z) = FU(z)$$

$$X(z) = (zI-E)^{-1} \cdot FU(z)$$

$$Y(k) = CX(k) \,/\, Z$$

$$Y(z) = CX(z)$$

$$Y(z) = CX(z) = C(zI-E)^{-1}FU(z)$$

$$G(z) = \frac{Y(z)}{U(z)} = C(zI-E)^{-1}F$$ – ф-ja переноса

$$G(z) = \begin{bmatrix} 7 & 0 \end{bmatrix} \left(\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 0.1891 & 0.0269 \\ -0.3764 & -0.0529 \end{bmatrix} \right)^{-1} \cdot \begin{bmatrix} 0.05729 \\ 0.02688 \end{bmatrix} =$$

$$= \begin{bmatrix} 7 & 0 \end{bmatrix} \begin{bmatrix} z-0.1891 & -0.0269 \\ 0.3764 & z+0.0529 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0.05729 \\ 0.02688 \end{bmatrix} =$$

$$= \begin{bmatrix} 7 & 0 \end{bmatrix} \cdot \frac{1}{(z-0.1891)(z+0.0529)} \begin{bmatrix} z+0.0529 & 0.0269 \\ -0.3764 & z-0.1891 \end{bmatrix} \cdot \begin{bmatrix} 0.05729 \\ 0.02688 \end{bmatrix}$$

$$\cancel{\frac{1}{z^2 - 0.1362z + 0.0001}}$$

$$= \frac{1}{z^2 - 0.1362z + 0.0001} \begin{bmatrix} 7z + 0.3703 & 0.1883 \end{bmatrix} \begin{bmatrix} 0.05729 \\ 0.02688 \end{bmatrix}$$

$$= \frac{0.40103z + 0.0212 + 0.0051}{z^2 - 0.1362z + 0.0001}$$

$$\Rightarrow \boxed{G(z) = \frac{0.40103z + 0.0263}{z^2 - 0.1362z + 0.0001}}$$ → ф-ja дискретної переноса система

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